Lista 2 CG

Augusto de Vis.

1- I padrão translação:

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+t_x \\ y+t_y \\ z+t_y \\ 1 \end{bmatrix} \xrightarrow{(4,8,0)} \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

II padrão rotação:

$$R_z = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & 0 \\ \sin\alpha & \cos\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha & 0 \\ 0 & \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_y = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

rotação em $z\left(\frac{\pi}{4}\right)$

$$\begin{bmatrix} \cos(45) & 0 & \sin(45) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(45) & 0 & \cos(45) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

III padrão escala

$$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x.s_x \\ y.s_y \\ z.s_z \\ 1 \end{bmatrix} \xrightarrow[75\%\ y]{200\%\ x}$$

3D

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0{,}75 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ ou } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0{,}75 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2D

IV padrão inversão

$$I_z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I_y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I_x = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$\Rightarrow$ reflexão X e Y :

3D $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ ou 2D $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(V) rotação 2D $\frac{\pi}{4}$ na origem (um terna de eixos) 1 deixar translada

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0{,}71 & -0{,}71 & 0 \\ 0{,}71 & 0{,}71 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

(VI) voltar pro centro $\Rightarrow$ rodar $\Rightarrow$ voltar $\Rightarrow$ transladar

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0{,}71 & -0{,}71 & 0 \\ 0{,}71 & 0{,}71 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

(VII) M = R órbita X TRANS X R própria

$$\begin{bmatrix} \cos\alpha & 0 & \sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & d \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\theta & 0 & \sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(2-) a) nos paralelogramos, as diagonais se encontram no centro do plano

$\Rightarrow \frac{p+q}{2} = \frac{s+r}{2} \Rightarrow s = p+q-r$. $\Rightarrow$ basta o X e Y de cada e achar Xs e Ys.

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b) Por padrão, o plano frontal é anti-horário.
⇝ Se considerarmos que y diz respeito a altitude, e que o triângulo é no plano xz, se o ponto 'e' tiver y positivo ele vê anti-horário (frente), se tiver negativo, vê atrás (horário).

3- a) $O_j = (0, 600)$ em m.
$\hat{x}_j = (1, 0)$ em m.
$\hat{y}_j = (0, -1)$ em m.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 600 \\ 0 & 0 & 1 \end{bmatrix}$$

b) $x: 100/800 = 1/8$
$y: 75/600 = 1/8$, mas tem que inverter

$x_m = x_j \cdot \frac{1}{8}$
$y_m = 75 - \frac{1}{8} y_j$

$$M_{jm} = \begin{bmatrix} 0,125 & 0 & 0 \\ 0 & -0,125 & 75 \\ 0 & 0 & 1 \end{bmatrix}$$

$O_j = (0; 75)$ m
$\hat{x}_j = (0,125; 0)$ m
$\hat{y}_j = (0; -0,125)$ m

$P_j = (200, 100) \Rightarrow P_m = (25; 62,5)$
$q_j = (400, 300) \Rightarrow q_m = (50; 37,5)$

$x_j = 8 x_m$
$y_j = 600 - 8 y_m$

$$M_{mj} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -8 & 600 \\ 0 & 0 & 1 \end{bmatrix}$$

⇝ $R_m = (4, 60)$
$\Rightarrow x_j = 8 \cdot 4 = 32$
$y_j = 600 - 8 \cdot 60 = 120$
$\Rightarrow R_j = (32, 120)$